Nighttime Activity and Recovery after the 2021 Henan Floods

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Abstract

The recovery of urban activity after a flood can vary across places and over time. I examine nighttime-light observations for Zhengzhou, Kaifeng, and Xinxiang between June and September 2021, combining quality flags, population weights, and historical county boundaries. The late-July median light indices are 0.779, 0.835, and 0.847 relative to each city's pre-event baseline. Their September values are 0.860, 0.845, and 0.939. Coverage checks reveal gaps during the strongest rainfall and show that threshold-based return dates depend on observation support. A separate radar experiment evaluates water recognition on labeled patches; a logistic model with local image features attains a test water intersection over union of 0.622. These results describe changes in observed lighting and performance in water recognition, with the economic interpretation constrained by the available measurements.

1 Introduction

Flood recovery becomes visible through everyday changes: lights return, journeys resume, and businesses reopen. These changes can occur at different speeds across a city. I wanted to understand whether satellite observations could reveal this uneven return to ordinary activity and how much evidence a recovery claim requires.

The July 2021 floods in Henan provide a relevant setting. NASA documented extensive surface flooding in northern Henan on July 26, following the severe mid-July rainfall [1]. Official accounts of the August recovery effort addressed housing, transport, farmland, public services, and business activity [2]. These accounts suggest that recovery has several dimensions. Nighttime radiance provides one observable signal associated with illuminated urban activity; its changes can reflect power availability and patterns of nighttime use.

The study asks how this signal changed after the storm, whether the subsequent paths differed across urban areas, and whether the observations support a sustained return to a chosen baseline level. A radar experiment examines how local image information improves water recognition. The two analyses address complementary measurements of disaster conditions and urban activity.

2 Data and study scope

The High-quality Daily Nighttime Light (HDNTL) dataset derives from NASA's VNP46A2 product and provides corrected radiance at 15 arc-seconds, approximately 500 m [3]. Its multiband GeoTIFF files identify dates in band descriptions and provide companion flags for reference values and interpolation. The study retains the published urban windows for the three cities and compares June–September in 2020 and 2021. These windows encompass urban cores and surrounding areas. Table 1 summarizes the 2021 observations after preprocessing.

Table 1: 2021 nighttime-light sample. Accepted dates meet the 70% coverage requirement. The bright mask uses pre-event information.

Urban window	Available dates	Accepted dates	Bright pixels
Zhengzhou	45	36	10,902
Kaifeng	44	38	1,370
Xinxiang	44	38	2,242

WorldPop provides estimated population counts for 2020 at 30 arc-seconds [4]. Historical county geometries from geoBoundaries represent 2017 boundaries [5]. The radar experiment draws 96 training, 32 validation, and 64 test patches from the HISEA-1 water dataset, retaining the published split membership [6]. The selected patches have 256×256 pixels and binary water labels. Their files provide insufficient information for assigning individual patches to the urban windows.

3 Measuring light variation

The preprocessing checks band dates, coordinate systems, radiance values, and flag categories. The HDNTL scale factor of 0.1 converts the stored values to radiance. The companion flag grids have different extents and dimensions from the radiance grids. Nearest-neighbour reprojection aligns these categorical flags to the light pixels. Pixels outside valid flag support remain excluded.

The baseline spans June 1–July 16, 2021. A pixel enters a fixed bright mask when it has at least five noninterpolated baseline observations and its median radiance reaches 5 nW cm^{-2} sr^{-1}. The same mask supports the 2020 comparison. Figure 1 shows the spatial concentration of baseline light, the retained bright areas, and the population context.

For bright mask M_c in urban window c, let V_{ct} contain its valid, noninterpolated pixels on date t. The daily mean and coverage are

$$L_{ct} = \frac{\sum_{i \in V_{ct}} r_{it}}{|V_{ct}|}, \qquad C_{ct} = \frac{|V_{ct}|}{|M_c|}. \tag{1}$$

The analysis accepts L_{ct} when $C_{ct} \geq 0.70$. It normalizes accepted values by their pre-event median,

$$I_{ct} = L_{ct} / \operatorname{median}_{s \in \mathcal{B}} L_{cs}, \tag{2}$$

where $\mathcal{B}$ denotes the baseline dates. The population-weighted version replaces the arithmetic mean with $\sum w_i r_{it} / \sum w_i$ over V_{ct} and uses its own baseline median. The weights w_i are bilinearly resampled WorldPop source-cell counts and indicate relative population concentration. The analysis does not sum these resampled counts to estimate exposed residents.

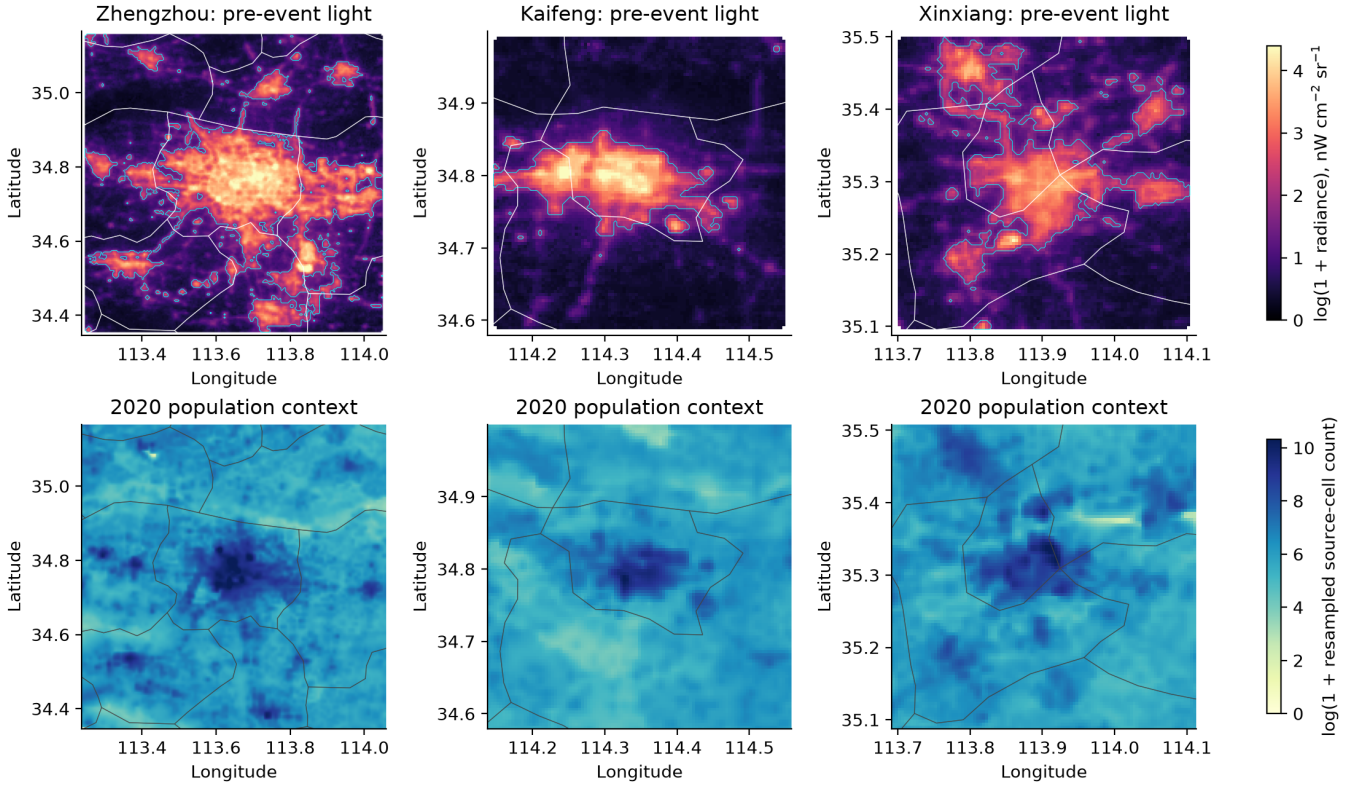


Figure 1: Baseline light and population context. Cyan contours mark the fixed bright masks. County outlines provide historical spatial context. The lower panels show transformed relative population weights.

4 Observation support and period comparisons

The available images cover 45 dates in Zhengzhou and 44 in each of Kaifeng and Xinxiang during June–September 2021. The coverage rule retains 36, 38, and 38 dates, respectively. Figure 2 shows that source availability and pixel coverage vary substantially through time. The three windows have no source bands for July 17–23. The first available dates after July 16 are July 24 in Zhengzhou, July 29 in Kaifeng, and July 26 in Xinxiang. The late-July observations therefore describe the aftermath of the strongest rainfall.

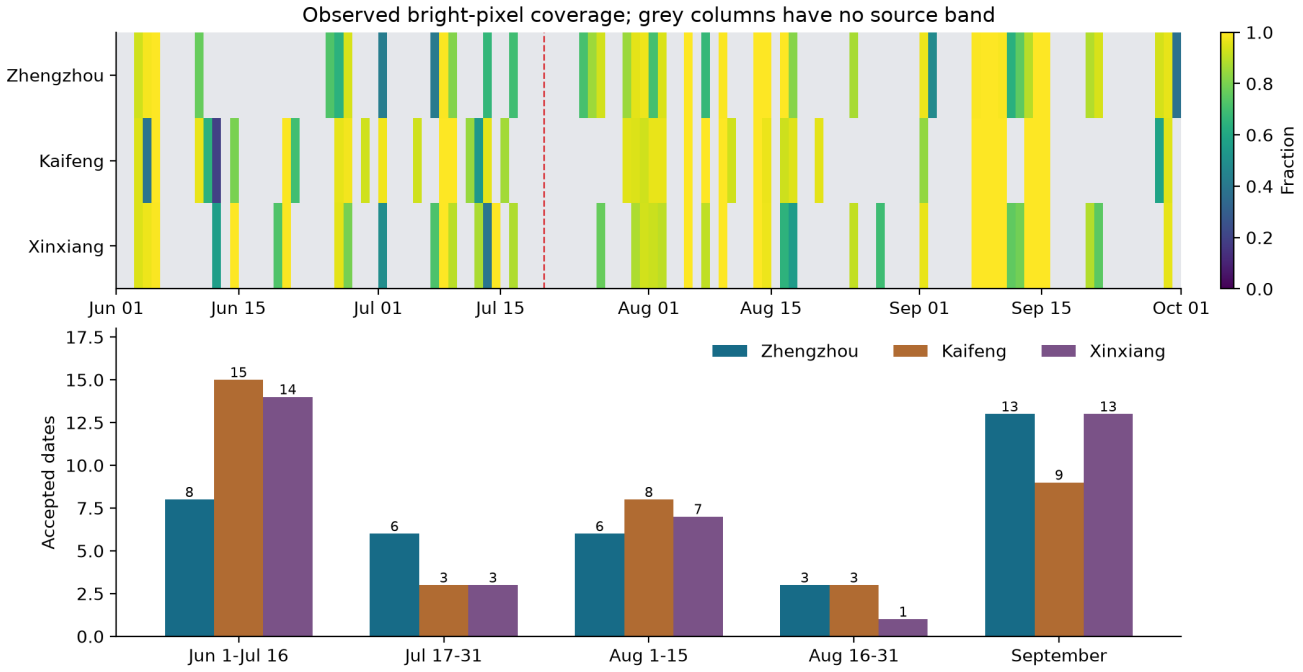


Figure 2: Temporal support for the light analysis. The upper panel shows noninterpolated coverage over fixed bright masks, with July 20 marked by a dashed line. The lower panel counts accepted dates within each comparison period.

Table 2: Median normalized light index by period. Each value summarizes accepted daily means; the baseline median equals one by construction.

Window	Pre-event	Jul 17–31	Aug 1–15	Aug 16–31	September
Zhengzhou	1.000	0.779	0.790	0.888	0.860
Kaifeng	1.000	0.835	0.840	0.782	0.845
Xinxiang	1.000	0.847	0.910	0.979	0.939

Table 2 shows lower late-July light levels in all three windows. Zhengzhou's median is 22.1% below its baseline, compared with 16.5% in Kaifeng and 15.3% in Xinxiang. Zhengzhou has six accepted late-July dates; the other windows each have three. These percentages describe observed radiance differences.

The subsequent paths differ. Zhengzhou's median rises from 0.779 in late July to 0.888 in the second half of August, then reaches 0.860 in September. Kaifeng remains below 0.85 across the later periods. Xinxiang reaches 0.939 in September. Its second-half August median of 0.979 rests on one accepted date, which gives that period much weaker support than its thirteen September observations.

5 Trajectories and supported return dates

Figure 3 retains the individual accepted observations and adds a trailing seven-calendar-day median. The median requires at least three accepted values within the window. This requirement leaves gaps when the data cannot support a local trajectory estimate.

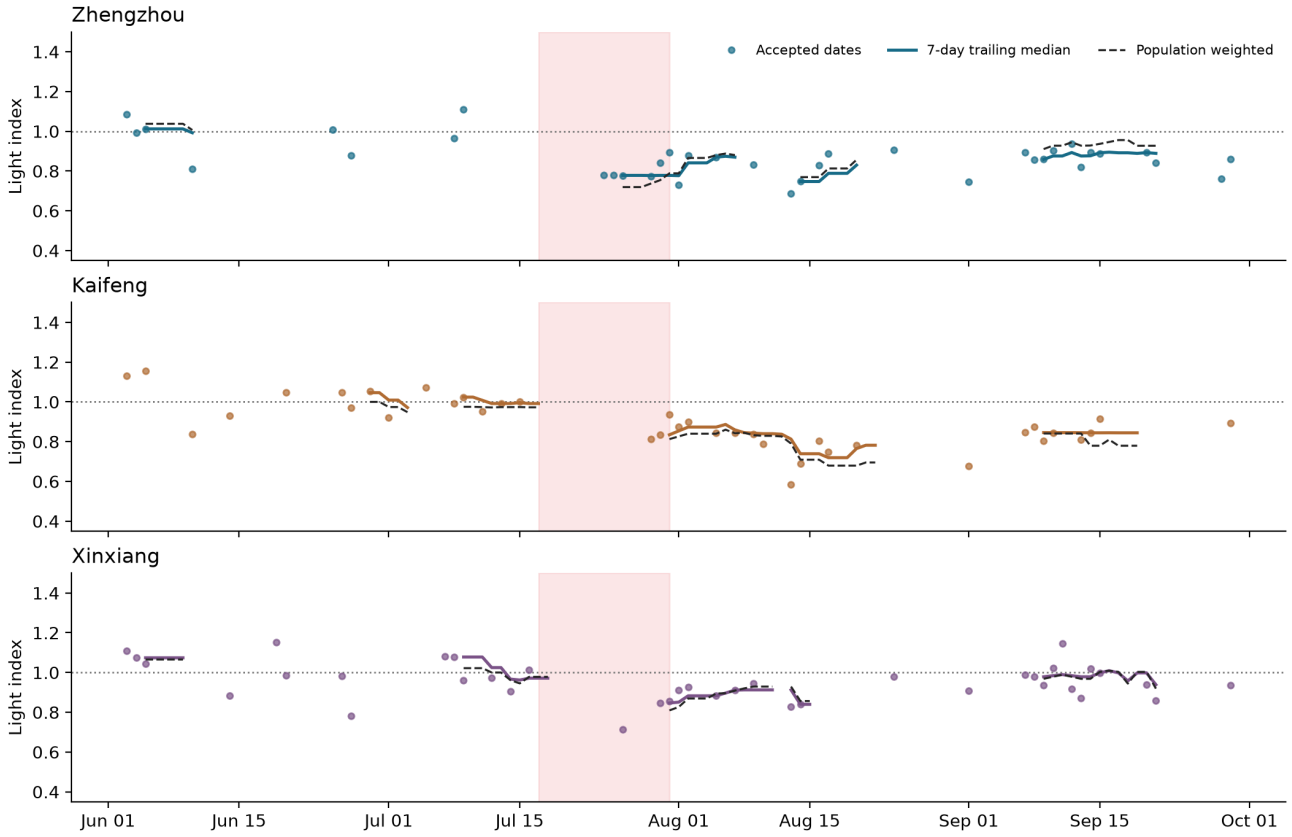


Figure 3: Normalized light trajectories. Points show accepted observations. Solid curves show supported trailing medians, and dashed curves use population weights. Shading marks July 17–31.

The population weighting changes the estimated level more visibly in some places and periods. Zhengzhou's September weighted median is 0.910, compared with 0.860 for the unweighted index. Kaifeng's corresponding values are 0.841 and 0.845, while Xinxiang's are 0.942 and 0.939. The Zhengzhou difference indicates that pixels with greater estimated population concentration carry a comparatively higher September light signal within the retained mask.

A supported return requires a smoothed July-window value below the chosen threshold, followed by three supported smoothed values at accepted observation dates above that threshold. Consecutive dates in this confirming sequence must be at most three days apart. Table 3 applies a threshold of 0.95 with the main coverage rule. Xinxiang's first qualifying sequence begins on September 9. Zhengzhou and Kaifeng have no qualifying sequence by September 30.

Table 3: Supported return to 95% of baseline with 70% minimum coverage.

Window	Result
Zhengzhou	Return not established by September 30
Kaifeng	Return not established by September 30
Xinxiang	First qualifying sequence begins September 9

This rule identifies a supported light-index crossing. The result leaves the actual timing of earlier changes unresolved, and the confirming sequence does not guarantee that every subsequent observation stays above the threshold.

6 Spatial variation and sensitivity

The spatial analysis retains county-window intersections with at least twenty bright pixels and normalizes each unit by its pre-event median. Figure 4 and Table 4 show September variation across seventeen units.

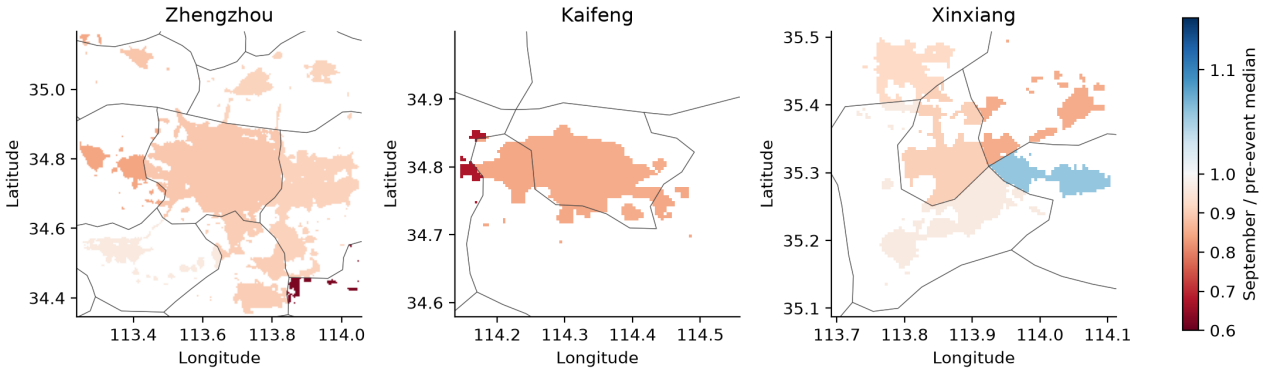


Figure 4: September light indices assigned to retained bright pixels within county-window intersections. White areas fall outside the retained bright units.

Table 4: September variation across county-window units.

Window	Units	Minimum index	Maximum index
Zhengzhou	9	0.623	0.961
Kaifeng	3	0.675	0.852
Xinxiang	5	0.852	1.058

Xinxiang's city-level value conceals units both below and above their local baselines. Zhengzhou has the widest range, including an index of 0.623. Each value describes the retained bright portion of a historical county within its image window.

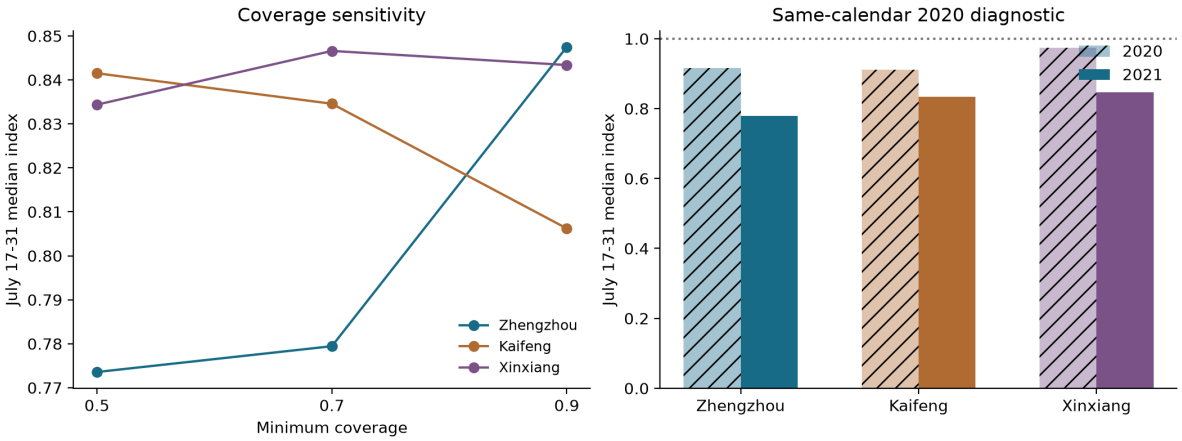


Figure 5: Sensitivity to coverage requirements and the same-calendar 2020 comparison. Each year uses its own pre-event median over the common 2021 bright mask.

Zhengzhou's late-July median rises from 0.774 at 50% coverage to 0.847 at 90%. Kaifeng spans 0.806–0.842 and Xinxiang 0.834–0.847. Xinxiang's September 9 crossing persists at 50% and 70% coverage for return thresholds of 0.90 and 0.95. Its 90% rule has insufficient July-window support, and a 1.00 return threshold yields no established return.

The 2020 late-July medians are 0.917, 0.911, and 0.974 for Zhengzhou, Kaifeng, and Xinxiang. All exceed their 2021 counterparts, while two remain below one. This diagnostic places the 2021 reductions within background calendar-period variation; year-specific conditions and uneven observation dates constrain causal interpretation.

7 Radar water recognition

The radar experiment uses labeled HISEA-1 patches from three 2021 flood events [6]. The small sample retains the source's training, validation, and test assignments. Each input is a grayscale PNG with a binary water mask. The model uses image intensities divided by 255; these values describe the distributed image representation.

The first baseline labels pixels with normalized intensity below 0.50 as water. A second baseline selects an intensity threshold from 0.02 to 0.98 in increments of 0.01 using validation intersection over union (IoU). The logistic model uses pixel intensity, a 3×3 local mean, a 3×3 local standard deviation, and a 9×9 mean. It standardizes these features and fits a class-balanced logistic classifier to 196,608 randomly sampled training pixels, with 2,048 pixels from each training patch. Validation data select a probability threshold from 0.10 to 0.90 in increments of 0.05.

The selected intensity and probability thresholds are 0.12 and 0.70. The test set contains 64 patches and 4,194,304 pixels. Precision, recall, F1, and IoU aggregate pixel counts across these patches. Here, $\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$, where TP, FP, and FN count true-positive, false-positive, and false-negative water pixels. The IoU interval resamples test patches 1,000 times. Table 5 gives the measured results.

Table 5: Test water-recognition performance. Intervals are 95% patch bootstrap intervals for IoU.

Method	Precision	Recall	F1	IoU	IoU interval
Intensity 0.50	0.235	0.996	0.380	0.234	[0.174, 0.298]
Tuned intensity	0.568	0.765	0.652	0.483	[0.397, 0.561]
Local logistic	0.754	0.780	0.767	0.622	[0.528, 0.708]

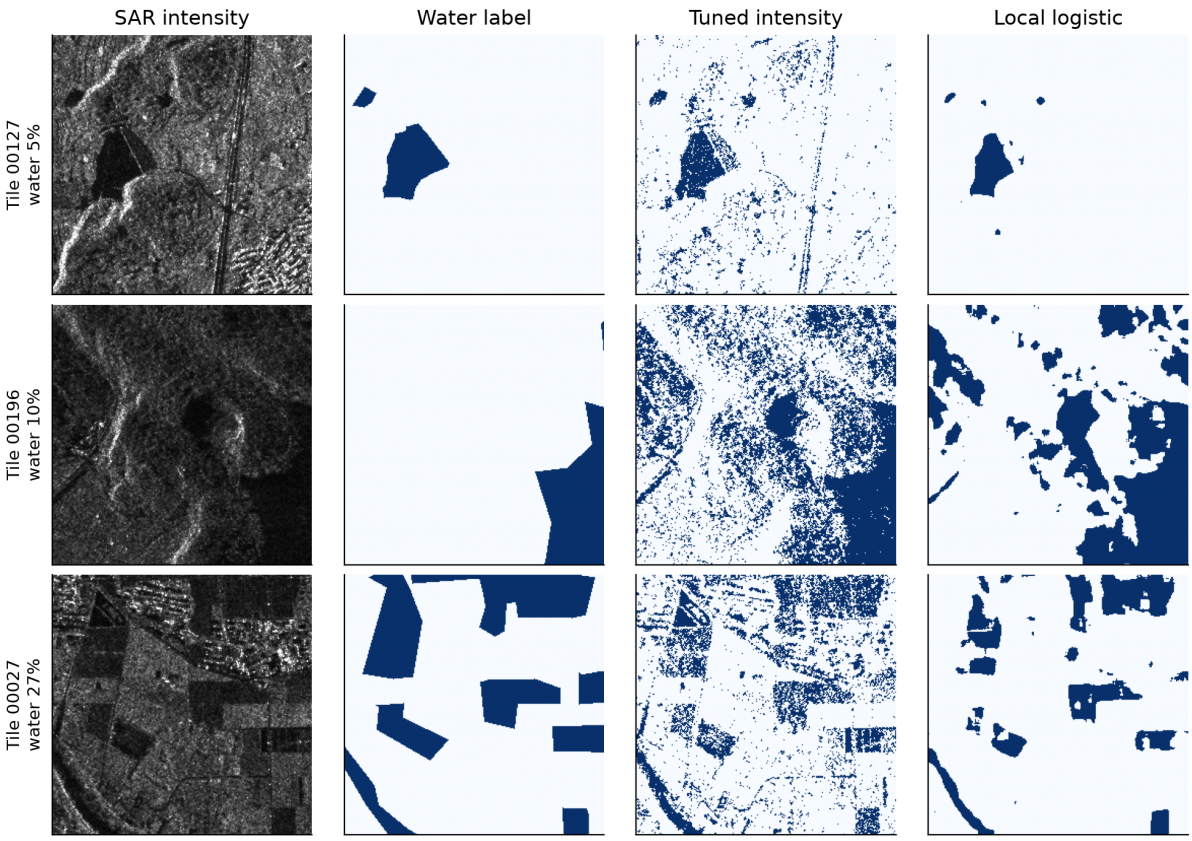


Figure 6: Test examples near the lower quartile, median, and upper quartile of labeled water fraction. Dark blue marks water. Example selection uses label coverage and does not use prediction scores.

8 Interpretation and conclusion

The radar results show a clear improvement from local image features. The fixed threshold has high recall but low precision, labeling many nonwater pixels as water. Validation tuning raises IoU from 0.234 to 0.483. The local logistic model reaches 0.622, an absolute increase of 0.139 over the tuned intensity baseline. Figure 6 illustrates how local information reduces scattered detections, while some dark regions remain difficult. Figure 7 summarizes the patch-level uncertainty.

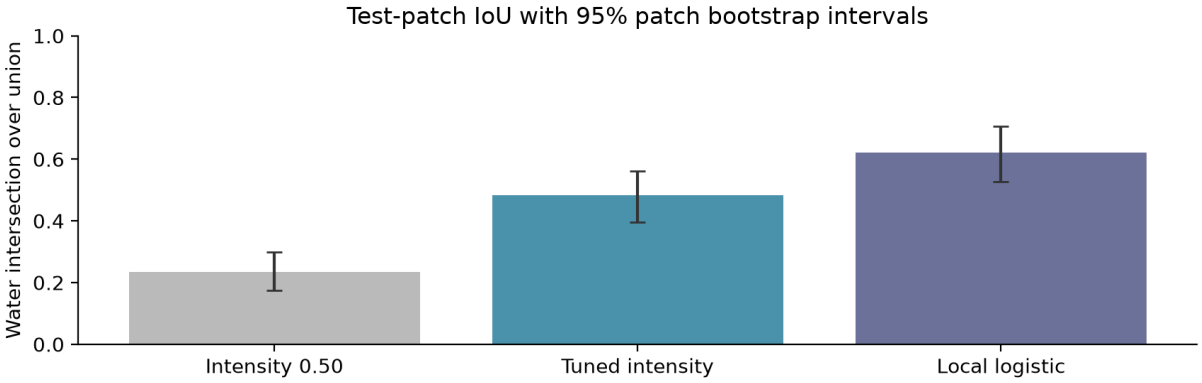


Figure 7: Aggregate test IoU and patch bootstrap intervals for the three water-recognition methods.

The radar labels identify water bodies, including features such as rivers, reservoirs, and paddy fields. Estimating newly inundated land would additionally require a spatially matched pre-event reference. The patch files provide no geographic coordinates or scene identifiers for that comparison. The source also randomly allocates patches across splits, so this evaluation supports performance on the sampled patches; it leaves generalization to independent flood scenes unmeasured. These properties prevent a direct spatial link between the radar predictions and the city light trajectories.

The urban analysis finds depressed late-July lighting followed by differing paths through September. Population weighting and county-window summaries show that the spatial composition of an index affects its interpretation. The coverage diagnostics reveal that a brighter observation can coexist with insufficient evidence for sustained return. The measurements also span an August period in which official recovery accounts discussed COVID-19 prevention alongside flood reconstruction [2]. The light trajectories therefore reflect an urban setting with several possible influences on activity. The available measurements provide insufficient information for attributing the changes solely to flooding or estimating economic losses.

The main finding is an uneven return toward pre-event lighting, with Xinxiang closer to its baseline in September than the other two windows. The accompanying sensitivity analysis shows how observation support constrains the timing of this return. The radar experiment establishes the value of local image statistics for recognizing water in the distributed patches. Together, the results connect satellite measurements to recovery questions while keeping each conclusion tied to the information its data can support.

References

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